The fontscale package

A flexible interface for setting font sizes

Oliver Beery

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1 Introduction

1.1 About

The fontscale package provides the following functionalities:

- Set font sizes using a classic or musical typographic scale (§2.1).
- Set arbitrary font sizes and font baselineskips for the standard LaTeX font size commands `\tiny`, `\scriptsize`, `\footnotesize`, `\small`, `\normalsize`, `\large`, `\Large`, `\LARGE`, `\huge`, and `\Huge` (§2.3).
- Defines lengths that store the current font size, font baselineskip, and the font size of `\normalsize` (§3.2).
- Set the font size relative to the current font size using more robust tools than the scalefnt and relsize packages (§3.3).
- Defines expl3 variables that store the font size and font baselineskip of each font size command from `\tiny` to `\Huge` and the current font size and font baselineskip (§§4.2–4.3).

1.2 Loading the package

Requirements:

- LaTeX 2_ε version 2023-11-01 or newer
- l3kernel version 2023-11-01 or newer
- fontscale is incompatible with the scalefnt package.

For historical reasons, the default Computer Modern font is available only in a number of discrete font sizes. If you get a warning that Computer Modern is not available in the requested size, you may need to add the following code before `\documentclass` to make Computer Modern available at arbitrary font sizes:

```
\RequirePackage{fix-cm}
```

Alternatively, you can use the Latin Modern font by loading the lmodern package.

The fontscale package has no package options. Instead, this package provides the command `\fontscalesetup{⟨key-value list⟩}` (§3.1) which sets the package keys (§2).

fontscale does not load or require any other packages.

When loaded, this package uses `\normalsize` after declaring and initializing the font size commands from `\tiny` to `\Huge`.

Many LaTeX classes have a font size option (e.g. `10pt`, `11pt`, `12pt`) which not only changes the set of font sizes, but also modifies additional settings such as the page layout and vertical spacing which were specifically designed to work with those font sizes. For this reason, you may want to set the class font size option close to the font size of `\normalsize` set by this package.

The font size commands from `\tiny` to `\Huge` defined by this package:

- First set `\@currsize` to have the same meaning as the font size command. This is needed only for compatibility.
- Set the font size and font baselineskip using `\fontsize` and `\selectfont`. Use `\fontscalesetup` (§3.1) to change the font sizes and font baselineskips set by the font size commands. The font baselineskip should not be confused with the paragraph baselineskip `\baselineskip`.
- Cannot be used in math mode.
- Have no other functionality. In particular, they do not change the vertical spacing for displayed math and list structures. This differs from the font size commands defined by many LaTeX classes. To add arbitrary code to the font size commands, use hooks, which are documented in lthooks and ltcmdhooks.

1.3 Syntax

This package defines some keys and commands that take as a value or argument a ⟨*floating point expression*⟩, ⟨*integer expression*⟩, ⟨*dimen expression*⟩, or ⟨*skip expression*⟩. This syntax has the same meaning as the arguments to `\fpeval`, `\inteval`, `\dimeval`, and `\skipeval`, respectively, which are documented in usrguide.

2 Keys

This section documents the keys provided by the fontscale package. This package has no package options. Set the package keys using `\fontscalesetup` `{`⟨*key-value list*⟩`}` (§3.1).

2.1 The key `typographic-scale`

The font sizes of the font size commands from `\tiny` to `\Huge` are initially set by the key `typographic-scale`.

```
typographic-scale = classic|musical
```

Choice key that sets the font size of each font size command from `\tiny` to `\Huge` using a classic or musical typographic scale. These are common methods of choosing a set of document font sizes. The initial value is `classic`.

Table 1: The font size of each font size command from `\tiny` to `\Huge` when setting the key `typographic-scale` = `classic` with different values for the choice key `classic/base`. The font sizes are in units of points. The "point" used depends on the value of the choice key `classic/point`.

font size command	10	11	12
`\tiny`	6	7	8
`\scriptsize`	7	8	9
`\footnotesize`	8	9	10
`\small`	9	10	11
`\normalsize`	10	11	12
`\large`	11	12	14
`\Large`	12	14	16
`\LARGE`	14	16	18
`\huge`	16	18	21
`\Huge`	18	21	24

```
classic
musical
```

Meta keys that set the key `typographic-scale` to the named value.

```
classic/base = 10|11|12
classic/point = pt|bp|dd|nd
```

Setting the key `typographic-scale` = `classic` sets the font size of each font size command from `\tiny` to `\Huge` using the classic typographic scale. The classic typographic scale consists of the traditional font sizes in points: 6, 7, 8, 9, 10, 11, 12, 14, 16, 18, 21, 24, 30, 36, 48, 60, and 72.[1] They have been used since the sixteenth century and are the default font sizes on most computer software.

The choice key `classic/base` sets the base font size of the classic typographic scale to the named value in units of points. The base font size is the font size of `\normalsize`. The font sizes of the other font size commands are the adjacent font sizes in the classic typographic scale in units of points. The initial value is `10`. The choice key `classic/point` sets which "point" is used. The initial value is `pt`. Table 1 displays the font size of each font size command from `\tiny` to `\Huge` when setting the key `typographic-scale` = `classic`.

1. See *The Elements of Typographic Style* by Robert Bringhurst.

Table 2: The font size of each font size command from `\tiny` to `\Huge` when setting the keys `typographic-scale` = `musical`, `musical/ratio` = 2, and `musical/notes` = 5 with different values for the key `musical/base`. The font sizes are in units of pt and rounded to 2 decimal places.

font size command	i	10 pt	11 pt	12 pt
`\tiny`	-4	5.74	6.32	6.89
`\scriptsize`	-3	6.60	7.26	7.92
`\footnotesize`	-2	7.58	8.34	9.09
`\small`	-1	8.71	9.58	10.45
`\normalsize`	0	10	11	12
`\large`	1	11.49	12.64	13.78
`\Large`	2	13.20	14.51	15.83
`\LARGE`	3	15.16	16.67	18.19
`\huge`	4	17.41	19.15	20.89
`\Huge`	5	20	22	24

```
musical/base = ⟨dimen expression⟩
musical/ratio = ⟨floating point expression⟩
musical/notes = ⟨integer expression⟩
```

Setting the key `typographic-scale` = `musical` calculates the font size of each font size command from `\tiny` to `\Huge` using the formula for the musical typographic scale:[2]

$$f_i = f_0 \times r^{i/n}$$

f_i is the font size of the ith note. f_0 is the base font size. n is the number of notes, the number of font sizes above f_0. r is the musical ratio, the ratio of the highest to the lowest note f_n/f_0.

The key `musical/base` sets the base font size f_0 to the result of evaluating the ⟨*dimen expression*⟩. The base font size is the font size of `\normalsize`. The initial value is 10 pt. The key `musical/ratio` sets the musical ratio r to the result of evaluating the ⟨*floating point expression*⟩. The initial value is 2. The key `musical/notes` sets the number of notes n to the result of evaluating the ⟨*integer expression*⟩. The initial value is 5. Table 2 displays the font size of each font size command from `\tiny` to `\Huge` when setting the key `typographic-scale` = `musical`.

2. I have referenced this article by Spencer Mortensen: `https://spencermortensen.com/articles/typographic-scale/`

2.2 The key `baselineskip-size-ratio`

The font baselineskips of the font size commands from `\tiny` to `\Huge` are initially set by the key `baselineskip-size-ratio`.

`baselineskip-size-ratio =` ⟨*floating point expression*⟩

Sets the font baselineskip of each font size command from `\tiny` to `\Huge` equal to its font size × the result of evaluating the ⟨*floating point expression*⟩. The initial value is 1.2. This key also affects the behavior of `\setfontsize` (§3.3).

2.3 Overwriting the previous keys

This subsection documents keys for setting the font sizes and font baselineskips of the font size commands from `\tiny` to `\Huge` in a more direct manner.

The user should take care to ensure that the lengths of the font sizes remain correctly ordered from `\tiny` to `\Huge`. This is important for typographic and syntactic consistency. If the font sizes are in the wrong order, then `\fontscalesetup` will issue a warning and some package features may not work correctly.

The syntax ⟨*font size command*⟩ represents the name of a font size command from `\tiny` to `\Huge`, omitting the backslash `\`.

⟨*font size command*⟩`/size-normalsize-ratio =` ⟨*floating point expression*⟩

Sets the font size of `\`⟨*font size command*⟩ equal to the font size of `\normalsize` × the result of evaluating the ⟨*floating point expression*⟩. Overwrites the font size set by the key `typographic-scale`. These keys are initially not set. The key `normalsize/size-normalsize-ratio` is not defined.

⟨*font size command*⟩`/size =` ⟨*dimen expression*⟩

Sets the font size of `\`⟨*font size command*⟩ to the result of evaluating the ⟨*dimen expression*⟩. Overwrites the font size set by the keys `typographic-scale` and ⟨*font size command*⟩`/size-normalsize-ratio`. These keys are initially not set.

⟨*font size command*⟩ `=` ⟨*dimen expression*⟩

Equivalent to ⟨*font size command*⟩`/size`=⟨*dimen expression*⟩.

⟨*font size command*⟩`/baselineskip-size-ratio =` ⟨*floating point expression*⟩

Sets the font baselineskip of `\`⟨*font size command*⟩ equal to its font size × the result of evaluating the ⟨*floating point expression*⟩. Overwrites the font baselineskip set by the key `baselineskip-size-ratio`. These keys are initially not set.

`⟨font size command⟩/baselineskip = ⟨skip expression⟩`

Sets the font baselineskip of `\⟨font size command⟩` to the result of evaluating the `⟨skip expression⟩`. Overwrites the font baselineskip set by the keys `baselineskip-size-ratio` and `⟨font size command⟩/baselineskip-size-ratio`. These keys are initially not set.

2.4 The key `magscale`

`magscale = ⟨floating point expression⟩`

Scales the font size and font baselineskip of each font size command from `\tiny` to `\Huge` by a factor equal to the result of evaluating the `⟨floating point expression⟩`. The new font baselineskips have no stretch and shrink components. This key is applied after all the other package keys. This key is initially not set.

3 Commands

This section documents the commands provided by the fontscale package.

3.1 Setting the keys

`\fontscalesetup ⟨*⟩ {⟨key-value list⟩}`

Sets and processes the fontscale package keys (§2) in `⟨key-value list⟩` and then uses `\normalsize`. Adding the optional star first resets all the package keys to their initial values. Can be used mid-document. The scope of the effect is local to the current group. Cannot be used in math mode.

3.2 Lengths

`\currentfontsize`

Dimen register (rigid length) that stores the current font size. This is more convenient than using `\f@size`.

`\currentfontbaselineskip`

Skip register (rubber length) that stores the current font baselineskip. This is more convenient than using `\f@baselineskip`. The font baselineskip should not be confused with the paragraph baselineskip `\baselineskip`.

`\currentnormalsize`

Dimen register (rigid length) that stores the current font size of `\normalsize`. This may be more convenient than using `\@ptsize`.

3.3 More font size commands

`\setfontsize [⟨skip expression⟩] {⟨dimen expression⟩}`

Sets the font size to the result of evaluating the ⟨*dimen expression*⟩. Sets the font baselineskip equal to the new font size × the value of the key `baselineskip-size-ratio` (§2.2). Adding the optional argument instead sets the font baselineskip to the result of evaluating the ⟨*skip expression*⟩. Cannot be used in math mode. This command is a more robust alternative to `\fontsize` + `\selectfont`.

`\scalefont {⟨floating point expression⟩}`

Scales the font size and font baselineskip by a factor equal to the result of evaluating the ⟨*floating point expression*⟩. The new font baselineskip has no stretch and shrink components. Cannot be used in math mode. This command is a more robust alternative to `\scalefont` from the scalefnt package.

`\stepfontsize {⟨integer expression⟩}`

Increases the font size by a number of "steps" equal to the result of evaluating the ⟨*integer expression*⟩. Changing the font size by one step means incrementing/decrementing from a font size equal to the font size of a font size command from `\tiny` to `\Huge` to the next larger/smaller \⟨*font size command*⟩ by using \⟨*font size command*⟩. Changing the font size by more than one step does not use any intermediate \⟨*font size command*⟩. If the result of evaluating the ⟨*integer expression*⟩ is zero and the current font size equals the font size of any \⟨*font size command*⟩, then `\stepfontsize` uses \⟨*font size command*⟩.

Some exceptions:

- Issues a warning if the current font size does not equal the font size of any font size command from `\tiny` to `\Huge`.
- Issues an error if the font size would be changed by more than nine steps.
- Issues a warning and uses `\tiny` if the new font size would be smaller than `\tiny`.
- Issues a warning and uses `\Huge` if the new font size would be larger than `\Huge`.
- Cannot be used in math mode.

3.4 Setting only the font baselineskip

`\setfontbaselineskip {⟨skip expression⟩}`

Sets the font baselineskip to the result of evaluating the ⟨*skip expression*⟩. Does not change the font size. Cannot be used in math mode.

3.5 Testing and debugging

`\printsampletext ⟨*⟩ {⟨text⟩}`

Prints ⟨*text*⟩ in each font size ordered from `\tiny` to `\Huge` each followed by `\par`. ⟨*text*⟩ can contain `\par` tokens. Adding the optional star reverses the order of the font sizes.

`\printfontsizecommand`

Tests if the current font size equals the font size of any font size command from `\tiny` to `\Huge`. If so, prints the name of that font size command. If not, prints "\undefined".

4 Programming

This section documents the expl3 programming support provided by the fontscale package.

4.1 Compatibility with `\text_purify:n`

This package uses `\text_declare_purify_equivalent:Nn` to make `\text_purify:n` correctly remove the text formatting commands defined by this package.

4.2 Variables set by `\fontscalesetup`

The syntax ⟨*font size command*⟩ represents the name of a font size command from `\tiny` to `\Huge`, omitting the backslash `\`.

`\l_fontscale_⟨font size command⟩_size_dim`

Stores the font size of \⟨*font size command*⟩. `\l_fontscale_normalsize_size_dim` is equivalent to `\currentnormalsize` (§3.2).

`\l_fontscale_⟨font size command⟩_baselineskip_skip`

Stores the font baselineskip of \⟨*font size command*⟩.

4.3 Variables set in the `selectfont` hook

`\l_fontscale_size_dim`

Stores the current font size. Equivalent to `\currentfontsize` (§3.2).

`\l_fontscale_baselineskip_skip`

Stores the current font baselineskip. Equivalent to `\currentfontbaselineskip` (§3.2).